ICAD RT-5 · Set 9 — Answer Sheet

Exam 16-11-2026 · 75 Q · 300 marks · 3 hours · +4 correct, −1 wrong, 0 blank

Name ..... Roll No ..... Date .....

Physics

1	a	b	c	d
2	a	b	c	d
3	a	b	c	d
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18	a	b	c	d
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Numerical — write the value

21	
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Chemistry

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18	a	b	c	d
19	a	b	c	d
20	a	b	c	d

Numerical — write the value

21	
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Mathematics

1	a	b	c	d
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3	a	b	c	d
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15	a	b	c	d
16	a	b	c	d
17	a	b	c	d
18	a	b	c	d
19	a	b	c	d
20	a	b	c	d

Numerical — write the value

21	
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23	
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Bubbles — single correct · Box — write the value · each subject out of 100

ICAD RT-5 · Set 9

Exam 16-11-2026 · 25 Physics · 25 Chemistry · 25 Mathematics · 300 marks · 3 hours · +4 / −1

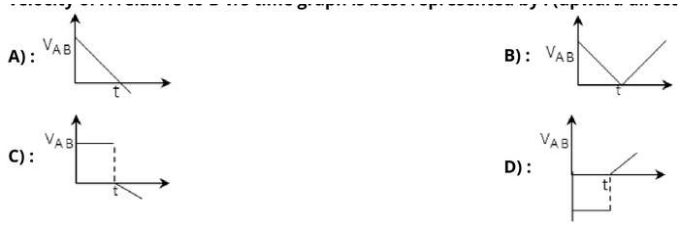
Physics

PART A · Q1–20 · SINGLE CORRECT

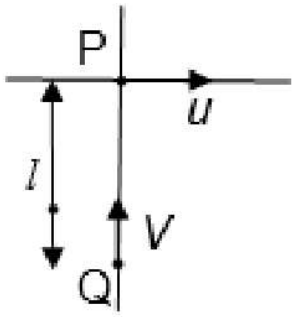
1. P4L1V3Q4 A toy cyclist completes one round of a square track of side 2 m in 40 seconds. What will be the displacement at the end of 3 minutes? (a) 52 m (b) Zero (c) 16 m (d) 2√2 m

2. P4L2V5Q7 Air friction causes vertical retardation equal to 10% of g (g = 10 ms⁻²). The maximum height will be decreased by: (a) 8% (b) 9% (c) 10% (d) 11%

3. P4XVCAT3Q10 A body A is thrown vertically upwards with such a velocity that it reaches a maximum height of h. Simultaneously another body B is dropped from height h. It strikes the ground and does not rebound. The velocity of A relative to B v/s time graph is best represented by: (upward direction is positive) (a) graph A (b) graph B (c) graph C (d) graph D — options are the v_AB–t graphs in the figure



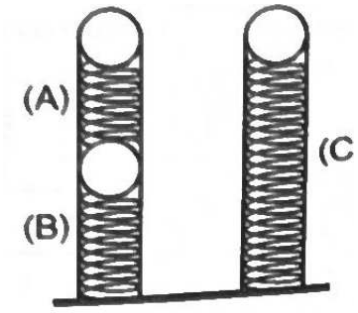
4. P4XVRT3Q2 A particle "P" moves with constant velocity u along the x-axis. Q always has velocity towards P. If Q moves with constant speed V then distance travelled by Q before meeting is _____ (a) uVl/√(V² − u²) (b) V²l/(V² − u²) (c) V²/(V² − u²) (d) lu/(V² − u²)



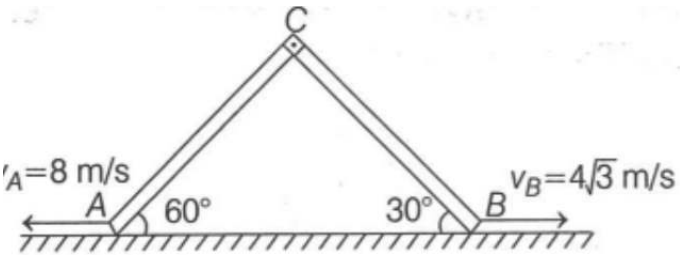
5. P5L3V3Q2 If τ = F × d and F = 50 N produces 100 Nm torque, what is the distance? What would be the force required (for the same torque) if the distance is halved? (a) 4 m, 100 N (b) 2 m, 100 N (c) 4 m, 200 N (d) 2 m, 50 N

6. P6L2V1Q4 When a car stops suddenly, the passengers are thrown forward. This is an example of which of the following? (a) Newton's first law (b) Newton's second law (c) Newton's third law (d) None of Newton's laws

7. P6L6V4Q5 Three springs, labeled (A), (B) and (C), are arranged as shown with three identical balls: on the left, spring (B) stands on the floor, a ball rests on it, spring (A) stands on that ball and a second ball rests on (A); on the right, spring (C) stands on the floor with the third ball on it. All the springs are parts cut from a single long spring such that A and B have equal natural length l but C has natural length 2l. All surfaces are smooth. Which spring is compressed by the smallest amount in equilibrium? (a) A (b) B (c) C (d) Cannot be determined with given information

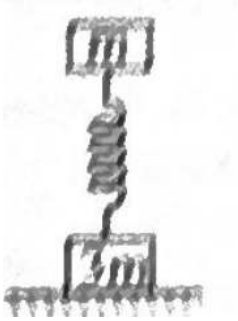


8. P6XVCAT4Q18 Two rods AC and BC are hinged at the end C and the ends A and B are moving along the surface with velocities v_A = 8 m/s (to the left) and v_B = 4√3 m/s (to the right), respectively as shown. AC makes 60° and BC makes 30° with the ground. The velocity of hinged point C is (a) √13 m/s (b) 2√13 m/s (c) 3√13 m/s (d) data insufficient



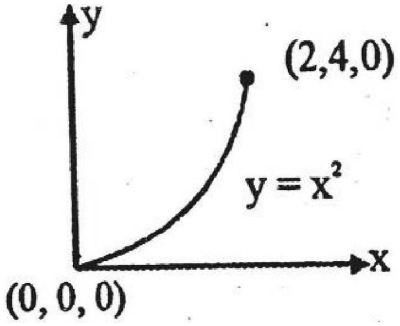
9. P7L4V5Q8 A smooth block is released at rest on a 45° incline and then slides a distance d. The time taken to slide is n times as much to slide on a rough incline than on a smooth incline. The coefficient of friction is: (a) μs = 1 − 1/n² (b) μs = √(1 − 1/n²) (c) μk = 1 − 1/n² (d) μk = √(1 − 1/n²)

10. P8L3V5Q9 Passage (Q9 & Q10): In the figure shown, the spring constant is k, the mass of the upper block is m and that of the lower block is 3m. The upper block is depressed down from its equilibrium position by a distance δ and released at t = 0. Q9. The minimum value of δ for the lower block loses contact with the ground is (a) 3mg/k (b) 2mg/k (c) 4mg/k (d) mg/k



11. P8L5V5Q3 Two blocks are connected to an ideal spring of stiffness 200 N/m. At a certain moment, the two blocks are moving in opposite directions (away from each other) with speeds 4 m/s and 6 m/s, and the instantaneous elongation of the spring is 10 cm. The rate at which the spring energy ($kx^2/2$) is increasing is (a) 500 J/s (b) 400 J/s (c) 200 J/s (d) 100 J/s

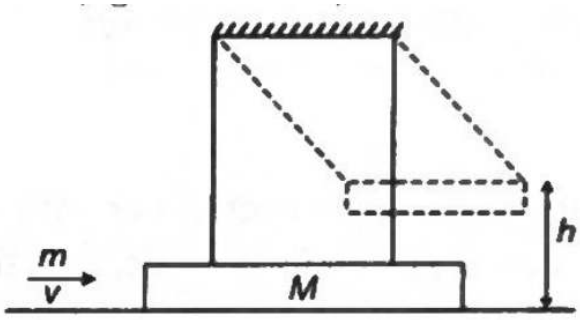
12. P8XVCAT5Q18 A force $F = (3xy - 5z)\,\hat{j} + 4z\,\hat{k}$ is applied on a particle. The work done by the force when the particle moves from the point (0, 0, 0) to the point (2, 4, 0) as shown in the figure is (a) 280/5 (b) 140/5 (c) 232/5 (d) 192/5



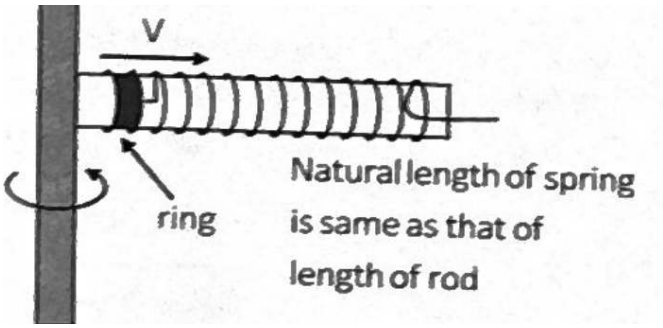
13. P9L1V5Q3 A skater of mass m standing on ice throws a stone of mass M with a velocity of v m/s in a horizontal direction. The distance over which the skater will move back (the coefficient of friction between the skater and the ice is μ) is: (a) $M^2v^2/(2m\mu g)$ (b) $Mv^2/(2m^2\mu g)$ (c) $M^2v^2/(2m^2\mu g)$ (d) $M^2v^2/(2m^2\mu^2 g)$

14. P9L3V5Q4 A particle of mass m moving with velocity v collides with a stationary particle of mass 2m. After collision, they stick together and continue to move together with velocity: (a) v/2 (b) v (c) v/4 (d) v/3

15. P9L3V5Q6 A large block of wood of mass M = 5.99 kg is hanging from two long massless cords. A bullet of mass m = 10 g is fired into the block and gets embedded in it. The system (block + bullet) then swings upwards, their centre of mass rising a vertical distance h = 9.8 cm before the (block + bullet) pendulum comes momentarily to rest at the end of its arc. The speed of the bullet just before collision is (Take g = 9.8 ms^{-2}): (a) 841.4 m/s (b) 811.4 m/s (c) 831.4 m/s (d) 821.4 m/s

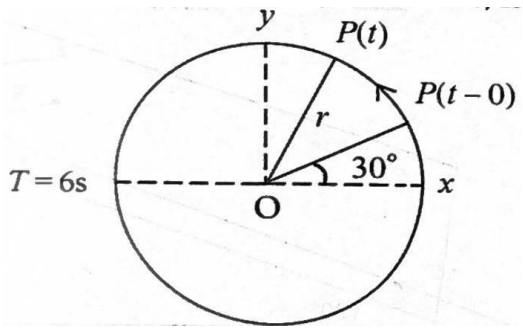


16. P10L3V5Q10 A ring attached with a spring is fitted in a smooth rod. The spring is fixed at the outer end of the rod. The mass of the ring is 3 kg & spring constant of spring is 300 N/m. The ring is given a velocity 'V' towards the outer end of the rod. And the rod is set to be rotating with an angular velocity ω. Then ring will move with constant speed with respect to the rod if: (Natural length of spring is same as that of length of rod.) (a) angular velocity of rod is increased continuously (b) ω = 10 rad/s (c) angular velocity of rod is decreased continuously (d) constant velocity of ring is not possible.



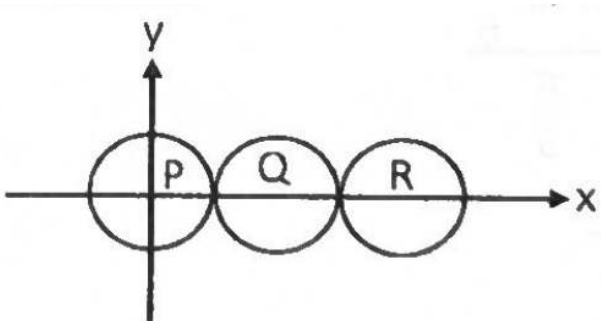
17. P10L5V3Q5 A 2.5 kg bucket has radius 1.5 m. Find minimum velocity at the top and corresponding normal force at the bottom. (a) 4.8 m/s and 100 N (b) 4.8 m/s and 130 N (c) 3.87 m/s and 150 N (d) 5 m/s and 90 N (The bob moves freely in the vertical circle; take g = 10 m/s^2.)

18. P10SBVPYQQ15 For particle P revolving round the centre O with radius of circular path r and regular velocity ω, as shown in the figure, the projection of OP on the x-axis at time t is (T = 6 s, P starts 30° above the x-axis): (a) x(t) = r cos(ωt − π/6) (b) x(t) = r cos(ωt + π/6) (c) x(t) = r sin(ωt + π/6) (d) x(t) = r cos(ωt)



19. P11L1V5Q4 Four particles of masses 2 kg, 3 kg, 5 kg and 8 kg are placed at the four corners of a square of side 2 m. The square lies in the XY-plane with 2 kg at origin and 3 kg on X-axis and 8 kg on Y-axis. Find the position of their centre of mass. (a) (8/9, 13/9) (b) (7/9, 11/9) (c) (11/9, 13/9) (d) (11/9, 8/9)

20. P11L2V2Q1 Three identical spheres, each of mass 1 kg are kept as shown in figure, touching each other, with their centres on a straight line. If their centres are marked P, Q, R respectively, the distance of centre of mass of the system from P is (a) (PQ + PR + QR)/3 (b) (PQ + PR)/3 (c) (PQ + QR)/3 (d) (PR + QR)/3

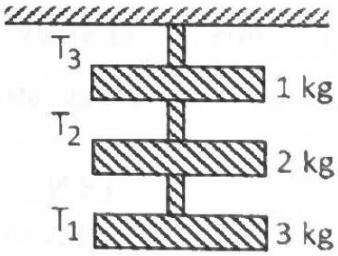


PART B · Q21–25 · NUMERICAL VALUE

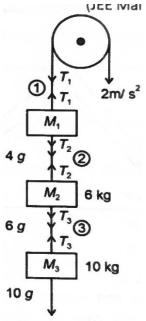
21. P3XVRT1Q24 If $P = 3\hat{i} + \sqrt{3}\hat{j} + 2\hat{k}$ and $Q = 4\hat{i} + \sqrt{3}\hat{j} + 2.5\hat{k}$, then the unit vector in the direction of P×Q is $(1/x)(\sqrt{3}\hat{i} + \hat{j} - 2\sqrt{3}\hat{k})$. The value of x is ____.

22. P5L2V5Q4 Three blocks of masses 1 kg, 2 kg and 3 kg hang one below the other from a ceiling (1 kg on top, 3 kg at the bottom) by light strings: T_3 is the tension in the string from the ceiling to the 1 kg block, T_2 between the 1 kg and 2 kg blocks, T_1 between the 2 kg and 3 kg blocks (see figure). Find T_3 in newton. ($g = 10\ m/s^2$)

T_3 in newton.

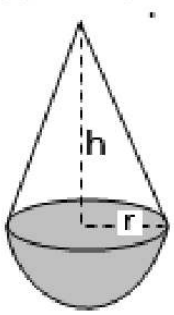


23. P6SBVPYQQ2 Three blocks M_1, M_2, M_3 having masses 4 kg, 6 kg and 10 kg respectively are hanging from a smooth pulley using ropes 1, 2 and 3 as shown in figure (rope 1 from the pulley to M_1, rope 2 from M_1 to M_2, rope 3 from M_2 to M_3). The tension in the rope 1, T_1, when they are moving upward with acceleration of 2 m s^{-2} is _____ N (if $g = 10\ m/s^2$).



24. P7SBVPYQQ14 A uniform chain of 6 m length is placed on a table such that a part of its length is hanging over the edge of the table. The system is at rest. The coefficient of static friction between the chain and the surface of the table is 0.5. The maximum length of the chain hanging from the table is ____ m.

25. P11XVCAT6Q21 A uniform solid right circular cone of base radius r is joined to a uniform solid hemisphere of radius r and of the same density, so as to have a common face. The centre of mass of the composite solid lies on the common face. The height of the cone is $r\sqrt{x}$. Find x



Chemistry

PART A · Q1–20 · SINGLE CORRECT

1. C1L6VSuggQ5 A gas has u_rms = 240 m/s at 150 K. If the gas is heated (at constant volume) to 600 K, what will be its new u_rms (in m/s)? (a) 480 m/s (b) 640 m/s (c) 960 m/s (d) 1200 m/s

2. C2L3V5Q8 Which of the following statements correctly explains why no photoelectric emission occurs when the frequency is below threshold, even at very high intensity? Statement I: The photons do not possess sufficient energy to overcome the work function. Statement II: Increasing intensity increases the number of photons but not their energy. Statement III: A larger number of low-energy photons can collectively eject one electron. (a) I and II only (b) II and III only (c) I only (d) I, II and III

3. C2L7V5Q2 What transition in the hydrogen spectrum would have the same wavelength as the Balmer transition n = 4 to n = 2 in the He^+ spectrum? (a) n = 4 to n = 2 (b) n = 3 to n = 2 (c) n = 3 to n = 1 (d) n = 2 to n = 1

4. C2L9VSuggQ5 How many electrons in the ground state of Fe (Z = 26) have both m_ℓ = 0 and s = +1/2? (Fe: [Ar] $4s^2\ 3d^6$; take the unpaired electrons as s = +1/2) (a) 5 (b) 6 (c) 7 (d) 8

5. C3L7VSuggQ3 30 g of sugar is dissolved to make 200 mL of solution. Calculate the mass by volume percent (w/v%). (a) 10% (b) 12.5% (c) 15% (d) 18%

6. C3L9VSuggQ3 The normality of 1% (mass/vol.) H_2SO_4 is nearly: (a) 0.02 (b) 0.2 (c) 0.1 (d) 1

7. C4L2V5Q4 A valence electron experiences a screening constant that is 2 units less than the number of inner electrons. If an atom has 20 inner electrons and its atomic number is 29, what is the effective nuclear charge (Z*) on the valence electron? (a) 7 (b) 9 (c) 11 (d) 17

8. C4L3V4Q5 Among the following ions what is the atomic number having highest ionic radius: Al^{3+}, Mg^{2+}, Na^+, O^{2-}, F^- (a) 8 (b) 9 (c) 11 (d) 13

9. C4XVCAT4Q42 S, T, U are aqueous chlorides of X, Y, Z (same period). U gives a white ppt with NaOH that dissolves in excess NaOH; T gives a white ppt that does not dissolve; S gives no ppt. Which statements are correct? (a) all three are metals (b) electronegativity decreases X→Y→Z (c) X, Y, Z could be Na, Mg, Al (d) first IE increases X→Y→Z (A) (a) & (c) (B) (a) & (b) (C) (b) & (c) (D) (c) & (d)

10. C5L1V5Q3 The bond between two identical non-metal atoms has a pair of electrons: (a) unequally shared between the two (b) equally shared between the two (c) transferred fully from one atom to another (d) none of these

11. C5L7V5Q8 Which combination is not allowed in the LCAO method (Z-axis as molecular axis)? (a) s + p_x (b) s + p_z (c) p_x + p_x (d) p_z + p_z

12. C5L8V5Q7 The electronic configuration of carbon is $1s^2 2s^2 2p^2$. There are 12 electrons in C_2. The correct electronic configuration of C_2 molecule is (a) $(\sigma 1s^2)(\sigma 1s^2)(\sigma 2s^2)(\sigma 2s^2)(\sigma 2p_z^2)(\pi 2p_x^2)$ (b) $(\sigma 1s^2)(\sigma 1s^2)(\sigma 2s^2)(\sigma 2s^2)(\pi 2p_x^2 = \pi 2p_y^2)$ (c) $(\sigma 1s^2)(\sigma 1s^2)(\sigma 2s^2)(\sigma 2s^2)(\sigma 2p_z^2)(\pi 2p_x^1 = \pi 2p_y^1)$ (d) $(\sigma 1s^2)(\sigma 1s^2)(\sigma 2s^2)(\sigma 2s^2)(\pi 2p_x^2 = \pi 2p_y^1)$

13. C5L9V3Q2 The dipole moment of a polyatomic molecule depends on: (a) Only the electronegativity difference of atoms (b) Only the number of polar bonds (c) Magnitude and spatial arrangement of individual bond moments (d) Only the molecular mass

14. C6L2VSuggQ4 A process is carried out infinitely slowly and maintains equilibrium at every stage. Which conclusion is correct? (a) The process must be isochoric. (b) The process must be natural and spontaneous. (c) The process is reversible and gives maximum work. (d) The process must involve no heat exchange.

15. C6L9V5Q8 Using the data provided, calculate the multiple bond energy (kJ mol −1) of a C≡C bond in C 2 H 2 . Given that the heat of formation of C 2 H 2 is 225 kJ mol −1 . (take the bond energy of a C—H as 350 kJ mol −1): 2C(s) → 2C(g); ΔH = 1410 kJ mol −1 H 2 (g) → 2H(g); ΔH = 330 kJ mol −1 (a) 1165 (b) 837 (c) 865 (d) 815

16. C6L11V4Q10 The enthalpy of fusion of water is 1.435 kcal/mol. The molar entropy change for the melting of ice at 0°C is: (a) 5.260 cal/(mol K) (b) 0.526 cal/(mol K) (c) 10.52 cal/(mol K) (d) 21.04 cal/(mol K)

17. C6L11VSuggQ1 It is impossible to attain the lowest temperature known as zero degree absolute. This is simple statement of: (a) first law of thermodynamics (b) second law of thermodynamics (c) third law of thermodynamics (d) none of the above

18. C6SBVPYQQ6 Match List-I (isothermal process for ideal gas system) with List-II (work done, $V_f > V_i$): (A) Reversible expansion; (B) Free expansion; (C) Irreversible expansion; (D) Irreversible compression. List-II: (I) w = 0; (II) $w = -nRT \ln(V_f/V_i)$; (III) $w = -p_{ex}(V_f - V_i)$; (IV) $w = -p_{ex}(V_i - V_f)$. (a) A-IV, B-I, C-III, D-II (b) A-IV, B-II, C-III, D-I (c) A-I, B-III, C-II, D-IV (d) A-II, B-I, C-III, D-IV

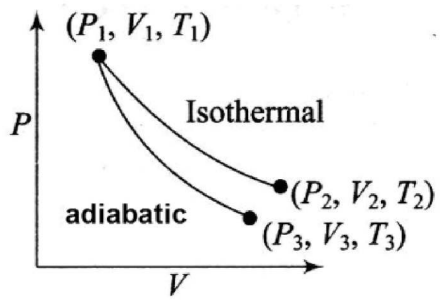
Match the List-I with List-II:

List-I (Isothermal process for ideal gas system)		List-II Work done ($V_f > V_i$)	
(A)	Reversible expansion	(I)	$w = 0$
(B)	Free expansion	(II)	$w = -nRT \ln \frac{V_f}{V_i}$
(C)	Irreversible expansion	(III)	$w = -p_{ex}(V_f - V_i)$
(D)	Irreversible compression	(IV)	$w = -p_{ex}(V_i - V_f)$

from the options given

19. C6SBVPYQQ23 S(g) + (3/2)O 2 (g) → SO 3 (g) + 2x kcal; SO 2 (g) + (1/2)O 2 (g) → SO 3 (g) + y kcal. The heat of formation of SO 2 (g) is given by: (a) x + y kcal (b) y − 2x kcal (c) (2x/y) kcal (d) 2x + y kcal

20. C6XVCAT6Q37 The reversible expansion of an ideal gas under adiabatic and isothermal conditions is shown in the figure (w = work done by the gas). Which of the following statement(s) is (are) correct ? (a) $T_1 = T_2$ (b) ΔS_isothermal > ΔS_adiabatic (c) w_isothermal > w_adiabatic (d) All of these



PART B · Q21–25 · NUMERICAL VALUE

21. C1XVRT1Q35 A mixture of two gases, A and B, diffuses through a pinhole in such a way that the fraction of A escaping per second is twice that of B. If M(A) = 16, determine M(B). (A) 32 (B) 64 (C) 128 (D) 256

22. C3XVRT2Q47 Mass of one atom of an element X is 2.32481 × 10 −23 g. Express this in amu.

23. C4SBVPYQQ43 If IUPAC name of an element is "Unununnium" then the element belongs to n th group of periodic table. The value of n is _____ .

24. C5XVCAT5Q47 Identify the Magnetic Nature. Given the following species: B 2 , C 2 , N 2 , O 2 , F 2 , find the number of paramagnetic species.

25. C6XVCAT6Q48 Cp − Cv for an ideal gas (per mole per kelvin) is _____ in calories (nearest integer).

Mathematics

PART A · Q1–20 · SINGLE CORRECT

1. M2L3VSuggQ3 If n(U) = 80, n(A) = 45, n(B) = 30 and n(A ∩ B) = 15, find n(A′ ∪ B′). (a) 50 (b) 65 (c) 35 (d) 80

2. M3L2VSuggQ3 If sin θ = 3/5, θ acute, then find cos(90° + θ). (a) −4/5 (b) 4/5 (c) 3/5 (d) −3/5

3. M4L4V2Q3 What is the range of y = 3 sin(x)? (a) [−1, 1] (b) [−3, 3] (c) [0, 3] (d) [−2, 2]

4. M4XVRT3Q59 If f : R → R be given by f(x) = 4^x/(4^x + 2) for all x ∈ R. Then, (a) f(x) = f(1 − x) (b) f(x) + f(1 − x) = 0 (c) f(x) + f(1 − x) = 1 (d) f(x) + f(x − 1) = 1

5. M6L3V4Q2 The solution set of the inequation |x + 2| ≤ 5 is: (a) (−7, 5) (b) [−7, 3] (c) [−5, 5] (d) (−7, 3)

6. M6SBVPYQQ6 The domain of the function f(x) = $\sin^{-1}[2x^2 - 3]$ + $\log_2(\log_(1/4)(x^2 - 5x + 5))$, where [t] is the greatest integer function, is: (a) (−√(5/2), (5 − √5)/2) (b) ((5 − √5)/2, (5 + √5)/2) (c) (1, (5 − √5)/2) (d) [1, (5 + √3)/2)

7. M7L1V6Q8 For b>0, b≠1 solve log_b(x) + log_x(b) = 5/2. (one or more options may be correct) (a) For b = 2, x = 4 solves (b) For b = 4, x = 2 solves (c) For every b > 1, x = b solves (d) For every b > 1, $x = b^2$ solves

8. M7L2V4Q10 For all positive integers n let f(n) = $\log_2002\ n^2$. Let N = f(11)+f(13)+f(14). Which relation is true? (a) 0 < N < 1 (b) N = 1 (c) 1 < N < 2 (d) N = 2

9. M8AL2V5Q9 If the angles of a quadrilateral are in A.P. whose common difference is 10°, then the angles of the quadrilateral are (a) 65°, 85°, 95°, 105° (b) 75°, 85°, 95°, 105° (c) 65°, 75°, 85°, 95° (d) 65°, 95°, 105°, 115°

10. M8ASBVPYQQ1 The common difference of the A.P. $a_1, a_2, \ldots\ldots, a_m$ is 13 more than the common difference of the A.P. $b_1, b_2, \ldots\ldots, b_n$. If $b_{21} = -277$, $b_{33} = -385$ and $a_{78} = 327$, then a_1 is equal to (a) 21 (b) 16 (c) 24 (d) 19

11. M8BL2VSuggQ2 Find S_3 for a = 2, r = 2: (a) 14 (b) 10 (c) 16 (d) 18

12. M8BL3V6Q5 The three numbers a, b, c between 2 and 18 such that (i) their sum is 25 (ii) the numbers 2, a, b are consecutive terms of an AP (iii) the numbers b, c, 18 are consecutive terms of GP (a) 5, 8, 12 (b) 5, 8, 10 (c) 8, 10, 12 (d) 4, 5, 8

13. M8CSBVPYQQ2 If in a triangle ABC the altitudes from the vertices A, B, C on the opposite sides are in H.P., then sin A, sin B, sin C are in (a) G.P. (b) Arithmetic–Geometric Progression (c) H.P. (d) A.P.

14. M8DL1VSuggQ1 What is the formula for the sum of the first n natural numbers? (a) n^2 (b) n(n+1)/2 (c) n(n−1)/2 (d) n(n+2)/2

15. M8DL3V5Q6 Find S n of the following sequence 3 + 5 + 9 + 15 + 23 + ... (a) $n(n^2+1)/3 + 3n$ (b) $n(n^2-1)/3 + 3n$ (c) $n(n^2+2)/3 + 3n$ (d) None of these

16. M8DL5V5Q6 If the H.M. of two numbers is 4, their A.M. 'A' and G.M. 'G' satisfy the relation $2A + G^2 = 27$. Find the numbers. (a) 6, 3 (b) 3, 4 (c) 4, 2 (d) 3, 5

17. M9L1VSuggQ1 What is the standard form of a complex number? (a) a + b (b) a + bi (c) ab + 1 (d) a + b + i

18. M9L3V4Q4 If $z = 4 - 3\sqrt{2}\,i$ then the value of $z\,\bar{z}$ = (a) 16 (b) 32 (c) 17 (d) 34

19. M9L3V5Q8 |1/(1 - i)^2 - 1/(1 + i)^2| = (a) 4 (b) 3 (c) 2 (d) 1

20. M9L6V2Q2 If a = cosθ + i sinθ, then (1 + a)/(1 − a) = (a) cot(θ/2) (b) cotθ (c) i cot(θ/2) (d) i tan(θ/2)

PART B · Q21–25 · NUMERICAL VALUE

21. M5AXVCAT3Q75 Find value of (tan 221/2° + tan 221/2°)/(1 − (tan 221/2°)²)

22. M5BXVRT2Q69 Let x = (Σ n=1 44 cos n°) / (Σ n=1 44 sin n°). Find the greatest integer that does not exceed 100x. (A) 141 (B) 241 (C) 341 (D) 441

23. M5CXVRT4Q71 Find the value of $2^{39}\sin(2^{11}\theta)\cos(2^{11}\theta)\cos(2^{12}\theta)\cos(2^{13}\theta)\ldots\cos(2^{45}\theta)$ where $\theta = \pi/2^{47}$.

24. M7XVCAT4Q71 If the solution set of (0.3)^(2x² − 3x + 6) < 0.00243 is (−∞, a) ∪ (a + 1/2, ∞) then value of 40a is

25. M9XVCAT6Q51 If (z + 2i)/(z − 2i) is purely imaginary, then |z| is (A) 1 (B) 2 (C) 1/2 (D) 1/4